

RUBBLEJACK · DESIGN DECISION RECORD

RubbleJack TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions D1 to D14, decided under Amish's 2026-10-03 pre-approval

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** decided. Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Each decision below is the recommendation made at the TRL 2 review and is decided as recommended.

■ Context

The scaffold (RBJ-PRC-001 v0.1) described a kit built around one commodity hand pump, with bolt-on spreader and cutter heads on standard cylinders and a lifting ram, and left open how the heads attach, which cylinder class to use, how strong each part must be and how the kit is carried. The TRL 2 review settled these so the TRL 3 calculations (RBJ-CAL-001), the model and the build plan could proceed. Choices that touch safety take the conservative option; each says what evidence would relax it.

Safety: RubbleJack lifts and spreads loads of up to 142 kN with oil at 700 bar, in collapsed structures. These decisions set the design basis on paper only. Nothing here makes the kit safe to use; it is not certified rescue equipment.

■ Decisions

Table 1. TRL 2 review decisions.

#	Decision	Reason	Affects
D1	Use the commodity 700 bar (10,000 psi) class throughout: a two-speed hand pump with internal relief at 700 bar, single-acting spring-return 15 t cylinders with 101 mm stroke, 700 bar hoses and couplers, and a gauge on the pump outlet	The most widely stocked class of hand hydraulics; two or more makers offer every part (R9); 142 kN per cylinder covers R1, R2 and R4	BOM lines 1 to 5; every calculation
D2	Give each tool its own cylinder (three identical cylinders) rather than one cylinder with quick-change heads	A tool change becomes one coupler off and one on, about 20 s (R8), with no thread to clean of dust under rubble; the cylinders stay interchangeable as spares. Cost: two more cylinders (USD 840)	BOM line 5; R8; value engineering
D3	Spreader: two jaw arms on one pivot pin, opened by a crosshead on the plunger and two links; 250 mm jaws; spreading only	Simple pinned linkage that local workshops can make; 49 to 64 kN at the tips and 262 mm opening (RBJ-CAL-001 section B). A single-acting cylinder gives no pulling or squeezing force, which is stated on the tool	Spreader parts; R2, R3
D4	Cutter: a guillotine shear with hardened S7 blades, single shear, rated for reinforcing bar up to 16 mm (B500B or ASTM A615 grade 60)	The cylinder can shear 18.7 mm bar at the upper strength, so a 16 mm rating keeps a ratio of 1.36. Conservative: no hardened bar, chain or larger bar. Relax only after proof cuts on CalRig at TRL 4	Cutter parts; R4
D5	Lifting ram: the cylinder stands in a locating ring on a 250 mm base plate; one 150 mm riser may go under it, never on top of the plunger; a tilting saddle on the plunger	Load always passes straight down through located steel; a riser on the plunger is the classic way a ram kicks out	Ram parts; R1, R12
D6	Materials: 4340 quenched and tempered arms and pins; S690QL links and cutter plates; S7 blades; 7075-T6 aluminium for bulky, lightly stressed parts	High strength where sections are small, aluminium where parts are bulky, so each carried load stays under 25 kg (R6)	Every made part; R6, R11
D7	Strength rule (new R11): every made load-bearing part has at least 1.5 on yield at the 700 bar relief pressure, on conservative hand calculations	Conservative safety basis while nothing is tested. Relax only with strain-gauged proof loads on CalRig	RBJ-CAL-001; R11
D8	Publish as an open engineering reference with a not-certified notice naming NFPA 1936, NFPA 1960 and EN 13204	Follows the patent and standards screen in RBJ-PRC-001	README, every document, tool markings
D9	Before any tool is used, it is proof-loaded on CalRig at 1.5 times the force at 700 bar and crack-checked; the plan for this is TRL 4 work	Conservative; matches the R1 verification method	First checks in RBJ-BLD-001; TRL 4
D10	Carry the kit as four loads in tool bags: pump, gauge and hoses; spreader; cutter; lifting ram set	Each stays under 25 kg packed (R6); four people can carry the whole kit	BOM line 23; R6
D11	The kit requires hardwood cribbing under the ram and next to every lifted load; cribbing is not designed or supplied (stays out of scope), and the documents give guidance on its use	Lifting without cribbing is the main hazard; local timber is widely available	Safety sections; R12
D12			

#	Decision	Reason	Affects
	Co-design: the first candidate partner to approach is a volunteer search and rescue association in southern Turkey (AKUT is the first candidate), with a local hydraulic workshop; first candidate region is the area hit in February 2023. Recorded as a first candidate, not agreed	A region with recent collapse experience, volunteer crews and a strong hydraulic trade	Co-design section of RBJ-PRB-001
D13	Shared blocks: the commodity hydraulics block (pump, hoses, couplers, gauge) is adopted as a shared block; CalRig is the proof-load rig; VoidScope stays a sibling	Re-use across the portfolio	RBJ-PRC-001
D14	The gauge stays fitted on the pump outlet whenever the kit is used	Crews see the pressure and stop before the relief valve is the only limit	BOM line 2; safety stops

■ Consequences

- The estimated cost rises above the USD 3,500 value-engineering target, mainly from D2 and D6 (see RBJ-DEC-001, Value engineering). Amish accepted cost overruns in the same instruction.
- R11 and R12 are added to RBJ-REQ-001.